Stats for Bart

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install necessary packages

Run this only once to have packages installed in your environment

```
install.packages("tidyverse", "glue", "effectsize", "wesanderson", "e1071")
```

load the packages

```
# General utility packages
library(tidyverse)
library(glue)

# Stats packages
library(effectsize)

# Plot pacakges
library(wesanderson)
pal <- wes_palette("Zissou1", 100, type = "continuous")

# SVM pacakge
library(e1071)
```

Load the tables

```
pre <- read_csv("assets/table/pre.csv") %>%
    mutate(ID = as.factor(ID)) %>%
    mutate(Group = as.factor(Group))

peri <- read_csv("assets/table/peri.csv") %>%
    mutate(ID = as.factor(ID)) %>%
    mutate(Group = as.factor(Group))

post <- read_csv("assets/table/post.csv") %>%
    mutate(ID = as.factor(ID)) %>%
    mutate(Group = as.factor(Group))
```

Making plots for pre conditon

```
g <- c()
g_lower <- c()
g_upper <- c()

df <- pre

for (i in 3:length(names(df))) {
    tmp <- hedges_g(flatten_dbl(df[, i]) ~ Group, data = df)
    g[i - 2] <- tmp$Hedges_g
    g_lower[i - 2] <- tmp$CI_low
    g_upper[i - 2] <- tmp$CI_high
}

sum_stat <- tibble(ROI = names(peri)[-c(1:2)], g = g, lower = g_lower, upper = g_upper)

plot <- sum_stat %>%
    mutate(ROI = fct_reorder(ROI, g, min)) %>%
    ggplot(aes(x = ROI, y = g, colour = g)) + geom_hline(yintercept = 0, linetype = "dashed") +
    geom_pointrange(aes(ymin = lower, ymax = upper)) + theme_minimal() + theme(text = element_text(size
    axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none") +
    scale_color_gradientn(colours = pal) + labs(y = "PTSD-like > resilient [Hedge's g]",
    title = "Pre condition", x = "")

ggsave(filename = "assets/figure/pre_g.png", plot = plot, device = "png", units = "mm",
    dpi = 300, width = 280, height = 120)
```

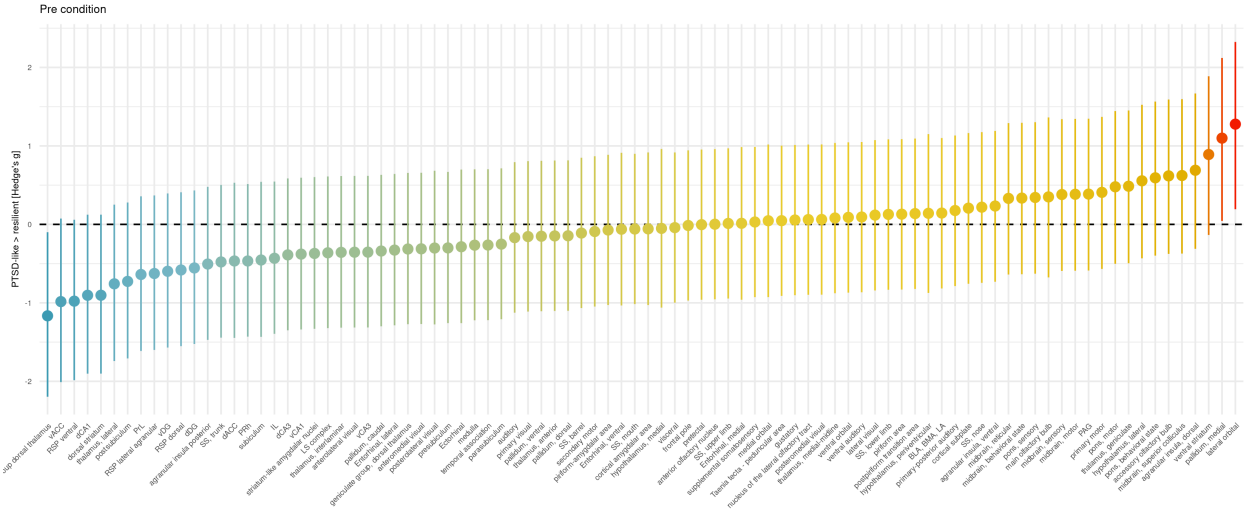


Figure 1: Figure 1

Making plots for peri conditon

```
g <- c()
g_lower <- c()
g_upper <- c()

df <- peri

for (i in 3:length(names(df))) {
    tmp <- hedges_g(flatten_dbl(df[, i]) ~ Group, data = df)
    g[i - 2] <- tmp$Hedges_g
    g_lower[i - 2] <- tmp$CI_low
    g_upper[i - 2] <- tmp$CI_high
}

sum_stat <- tibble(ROI = names(peri)[-c(1:2)], g = g, lower = g_lower, upper = g_upper)

plot <- sum_stat %>%
    mutate(ROI = fct_reorder(ROI, g, min)) %>%
    ggplot(aes(x = ROI, y = g, colour = g)) + geom_hline(yintercept = 0, linetype = "dashed") +
    geom_pointrange(aes(ymin = lower, ymax = upper)) + theme_minimal() + theme(text = element_text(size
    axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none") +
    scale_color_gradientn(colours = pal) + labs(y = "PTSD-like > resilient [Hedge's g]",
    title = "Peri condition", x = "")

ggsave(filename = "assets/figure/peri_g.png", plot = plot, device = "png", units = "mm",
    dpi = 300, width = 280, height = 120)
```

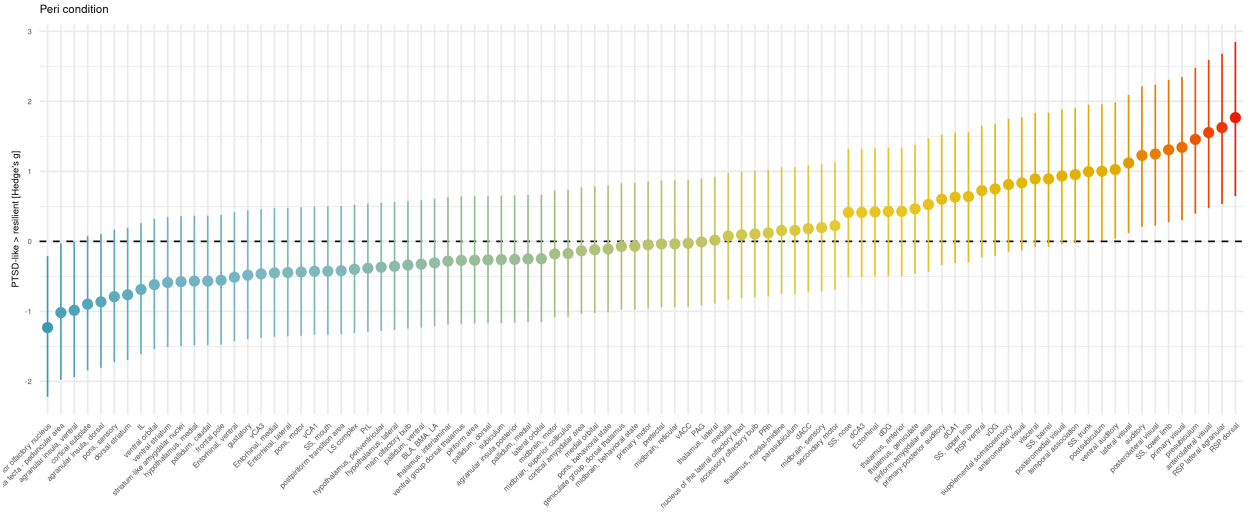


Figure 2: Figure 2

Making plots for post conditon

```
g <- c()
g_lower <- c()
g_upper <- c()

df <- post

for (i in 3:length(names(df))) {
    tmp <- hedges_g(flatten_dbl(df[, i]) ~ Group, data = df)
    g[i - 2] <- tmp$Hedges_g
    g_lower[i - 2] <- tmp$CI_low
    g_upper[i - 2] <- tmp$CI_high
}

sum_stat <- tibble(ROI = names(peri)[-c(1:2)], g = g, lower = g_lower, upper = g_upper)

plot <- sum_stat %>%
    mutate(ROI = fct_reorder(ROI, g, min)) %>%
    ggplot(aes(x = ROI, y = g, colour = g)) + geom_hline(yintercept = 0, linetype = "dashed") +
    geom_pointrange(aes(ymin = lower, ymax = upper)) + theme_minimal() + theme(text = element_text(size
    axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none") +
    scale_color_gradientn(colours = pal) + labs(y = "PTSD-like > resilient [Hedge's g]",
    title = "Post condition", x = "")

ggsave(filename = "assets/figure/post_g.png", plot = plot, device = "png", units = "mm",
    dpi = 300, width = 280, height = 120)
```

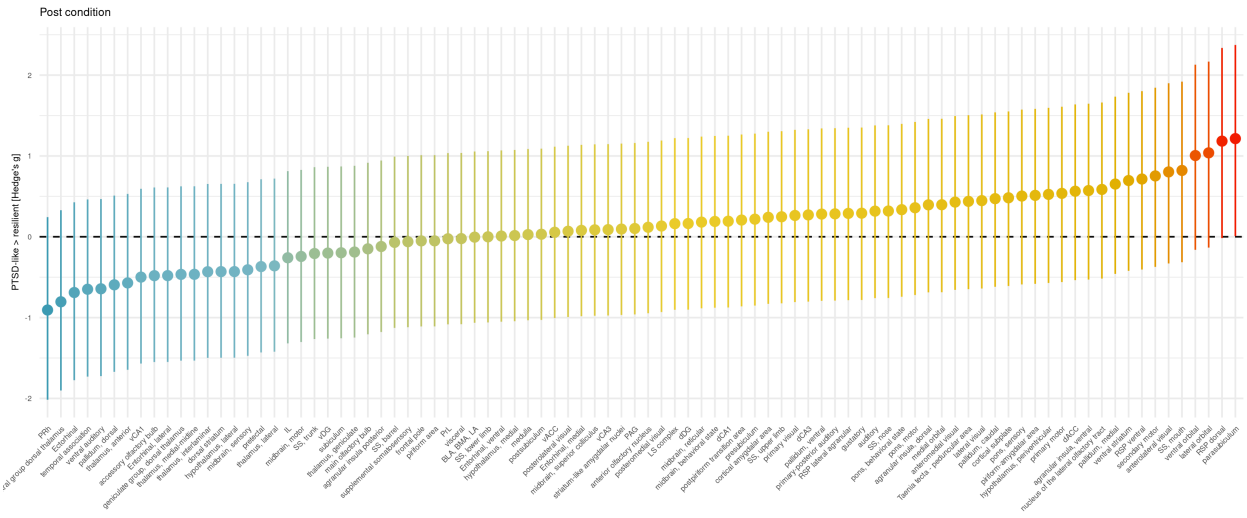


Figure 3: Figure 3

Support vector machine.

Following the guide from https://www.datacamp.com/community/tutorials/support-vector-machines-r

```r
x <- as.matrix(post[, -c(1:2)])
y <- as.factor(post$Group)

dat = data.frame(x, y = y)
svmfit = svm(y ~ ., data = dat, kernel = "radial", cost = 5, scale = FALSE)
print(svmfit)

post.predict <- predict(svmfit, dat)

detection <- mean(levels(post.predict)[post.predict] == levels(dat$y)[dat$y])
print(glue("Prediction of Post using Post is {detection*100} %"))
```

```
##
## Call:
## svm(formula = y ~ ., data = dat, kernel = "radial", cost = 5, scale = FALSE)
##
##
## Parameters:
##    SVM-Type:  C-classification
##  SVM-Kernel:  radial
##        cost:  5
##
## Number of Support Vectors:  12
##
## Prediction of Post using Post is 100 %
```

```r
x <- as.matrix(pre[, -c(1:2)])
y <- as.factor(pre$Group)

dat2 = data.frame(x, y = y)


pre.predict <- predict(svmfit, dat2)

detection <- mean(levels(pre.predict)[pre.predict] == levels(dat2$y)[dat2$y])
print(glue("Prediction of Pre using Post is {detection*100} %"))
```

```
## Prediction of Pre using Post is 62.5 %
```

Prediction isn't really above change level ($\sim 50\%$)

```r
x <- as.matrix(peri[, -c(1:2)])
y <- peri$Group


dat3 = data.frame(x)


peri.predict <- predict(svmfit, dat3)

# tibble(truth = y, predicted = peri.predict) %>% kable('pipe')
```

```
detection <- mean(levels(peri.predict)[peri.predict] == levels(dat3$y)[dat3$y])
print(glue("Prediction of Peri using Post is {detection*100} %"))
```

```
## Prediction of Peri using Post is NaN %
```

Prediction isn't really above change level (~ 50%)